

Software Requirements Specification (SRS) for Internship Progress Dashboard

1. Introduction

Purpose of the System

The primary purpose of the Internship Progress Dashboard is to provide a dedicated, web-based platform for managing, tracking, and monitoring the progress of interns within an organization. This system aims to streamline task assignment, enhance collaboration between interns and mentors, and provide clear, real-time visibility into an intern's performance and overall project status.

Scope of the Project

The project scope encompasses the design, development, and deployment of a full-stack web application supporting two main user roles: Intern (User) and Mentor. The system will provide core functionalities including user authentication, task management via a Kanban board, detailed progress tracking, and performance analytics.

Definitions

- **Intern:** A user with basic technical knowledge who uses the system to view, manage, and update the status of assigned tasks.
- **Mentor:** A user responsible for assigning tasks, monitoring the progress of assigned interns, and accessing performance analytics.
- **Task:** A defined unit of work assigned to an Intern, possessing attributes like a title, description, and status.
- **Progress:** The current state of an assigned Task, measured by its status (e.g., To Do, In Progress, Completed).
- **Kanban Board:** A visual workflow tool used to display tasks in columns based on their status, allowing for drag-and-drop interaction.

2. Overall Description

2.1 Product Perspective

The Internship Progress Dashboard is a standalone, new web-based dashboard system.

- **Frontend:** A responsive web interface for user interaction (Interns and Mentors).
- **Backend:** A server-side application to handle business logic, data processing, and API endpoints.
- **Database:** A persistent data store for all system information (user data, tasks, progress logs).

2.2 User Classes

The system defines two primary user classes with distinct access levels and functionalities:

User Class	Description
Intern (User)	Individuals who are assigned tasks and responsible for updating their status.
Mentor	Individuals responsible for task assignment, monitoring, and performance review of Interns.

2.3 Operating Environment

The system is designed to operate in the following environments:

- **Web Browser:** Modern web browsers, including Google Chrome, Microsoft Edge, and Mozilla Firefox.
- **Device:** Must be fully responsive and functional on both desktop computers and mobile devices (tablets and smartphones).

2.4 Constraints

- **Internet Requirement:** Continuous internet connectivity is mandatory for all system access and operations.
- **Authentication Mandatory:** All users must successfully authenticate (login) before accessing any system features.

3. 👥 User Characteristics

Intern

- **Technical Knowledge:** Basic understanding of web applications.
- **UI Preference:** Requires a simple, intuitive, and distraction-free user interface.
- **Focus:** Primarily focused on efficient task management and progress reporting.

Mentor

- **Technical Knowledge:** Moderate technical proficiency.
- **Needs:** Requires robust tools for monitoring, control, and data visualization (analytics).

4. ⚙️ Functional Requirements

🔒 Authentication

ID	Description
FR-Auth-01	The system shall allow new users to register with their name, email, and password.
FR-Auth-02	The system shall allow registered users to log in using their email and password.
FR-Auth-03	Authentication shall be secured using JSON Web Tokens (JWT).
FR-Auth-04	The system shall provide a secure logout mechanism, invalidating the current user session.

👤 Intern (User) Functionalities

- **FR-I-01:** The Intern shall be able to view a list of all tasks assigned to them.
- **FR-I-02:** The Intern shall be able to update the status of a task through predefined transitions (To Do → In Progress → Completed).
- **FR-I-03:** The Intern shall be able to interact with tasks on the Kanban Board using drag-and-drop functionality to change the task status.
- **FR-I-04:** The Intern shall be able to view a personal progress dashboard summarizing their task completion rate and current workload.
- **FR-I-05:** The Intern shall be able to edit and update their profile information.

Mentor Functionalities

- **FR-M-01:** The Mentor shall be able to create and assign new tasks to specific Interns.
- **FR-M-02:** The Mentor shall be able to view a list of all currently managed Interns.
- **FR-M-03:** The Mentor shall be able to monitor the real-time progress of all tasks assigned to all their Interns.
- **FR-M-04:** The Mentor shall be able to view analytics, including overall completion rates and performance statistics for individual interns and the group.
- **FR-M-05:** The Mentor shall be able to manage the assignment of Interns to specific tasks or projects.

Dashboard

- **FR-D-01:** The dashboard shall display key performance indicators (KPIs) such as total tasks, tasks in progress, and the overall completion rate.
- **FR-D-02:** The dashboard shall provide overview cards summarizing critical information relevant to the logged-in user's role (e.g., pending tasks for Intern, lagging interns for Mentor).

Kanban Board

- **FR-K-01:** The Kanban board shall display tasks categorized under three columns: To Do, In Progress, and Completed.
- **FR-K-02:** Tasks displayed on the board shall be interactive, supporting the drag-and-drop movement between status columns.

Progress Tracking

- **FR-P-01:** The system shall display progress using appropriate visualizations (graphs or charts).
- **FR-P-02:** The system shall provide insights into task completion trends and bottlenecks.

5. **Non-Functional Requirements**

Performance

- **NFR-P-01:** All main dashboard and task list pages shall load in under 2 seconds under normal network conditions.
- **NFR-P-02:** Real-time updates (e.g., task status changes) shall be reflected on all relevant screens within 1 second.

Security

- **NFR-S-01:** All passwords shall be hashed before being stored in the database.
- **NFR-S-02:** All user sessions shall be secured using JWT-based authentication.
- **NFR-S-03:** Access to Mentor functionalities shall be strictly enforced via role-based authorization.

Usability

- **NFR-U-01:** The user interface (UI) shall be simple, clean, and minimize cognitive load.
- **NFR-U-02:** Navigation between core sections (Dashboard, Tasks, Analytics) shall be intuitive and easily accessible.

Scalability

- **NFR-L-01:** The system architecture shall be capable of handling an increasing number of Interns and Mentors (e.g., up to 500 users) without significant degradation in performance.

Availability

- **NFR-A-01:** The system shall maintain an availability (uptime) of 99%.

6. System Features Summary

The Internship Progress Dashboard will be centered around three main features:

- **F-01: Task Management System:** Core functionality allowing for task creation, assignment, status updates, and visual management via a Kanban board.
- **F-02: Real-time Progress Tracking:** Provides up-to-date visualization and reporting of task status and overall intern performance.
- **F-03: Role-based Access:** Ensures strict separation of access and functionalities between the Intern and Mentor roles.

7. Data Requirements

Data Entity	Required Attributes
User Data	Name, Email (Unique), Password (Hashed), Role (Intern/Mentor)
Task Data	Title, Description, Status (To Do, In Progress, Completed), Priority, Assigned Intern ID, Mentor ID, Creation Date, Due Date

Data Entity	Required Attributes
Progress Data	Task ID, Timestamp of Status Change, Old Status, New Status

8. Security Requirements

- **SR-01: Secure Login:** All login attempts must be validated against hashed passwords, and communication must use HTTPS.
- **SR-02: Role-based Authorization:** The system must restrict access to Mentor-specific APIs and data (e.g., assignment, analytics) to only authenticated Mentor users.
- **SR-03: Data Protection:** Sensitive user data (e.g., email) must be protected in the database, and user passwords must never be stored in plain text.

9. Error Handling

- **EH-01: Invalid Login:** The system shall display a clear, non-specific error message for invalid login attempts (e.g., "Invalid email or password").
- **EH-02: Unauthorized Access:** Any attempt to access restricted resources shall result in an immediate "403 Forbidden" error or a similar appropriate message.
- **EH-03: Empty States:** The UI shall display informative messages when data lists are empty (e.g., "No tasks assigned yet").

10. Assumptions & Dependencies

- **Assumption 1:** It is assumed that all end-users (Interns and Mentors) have reliable and consistent internet access to use the platform.
- **Dependency 1:** The successful operation of the front-end is dependent on the availability and stability of the back-end API.
- **Dependency 2:** A working email system is assumed for user registration verification or password resets.

11. Future Scope

The following features are not part of the initial release but are planned for future development:

- **FS-01: Notifications System:** Implementation of email or in-app notifications for task assignments, due date reminders, and status changes.
- **FS-02: AI-based Task Suggestions:** Introducing a system that suggests tasks or next steps to Interns based on their past progress and current project goals.

- **FS-03: Chat System:** Integrating a basic real-time chat functionality for direct communication between an Intern and their assigned Mentor.